

Scale Reduction for Compactly Supported Polynomial Kernels

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Less psychosis at the expense of a life devoid of positive emotions.

Abstract

This paper presents the second generation of the author's Edge Vector Theory program. In contrast to the original framework, the present work is deliberately restricted to standard mathematical analysis. Its purpose is to isolate a mathematically self-contained analytical component, derive its structural identities, and establish its fundamental properties independently of the original conceptual framework.

1 Definitions

For

$$\rho_k(x) = \frac{k+1}{2}(1-|x|)_+^k,$$

define

$$\rho_{k,a}(x) = \frac{1}{a}\rho_k(x/a),$$

and

$$F_k(a, b) = \int_{\mathbb{R}} \rho_{k,a}(x) \rho_{k,b}(x) dx.$$

2 Scaling

For $0 < a \leq b$ and $r = a/b$,

$$F_k(a, b) = \frac{1}{b}H_k(r),$$

where

$$H_k(r) = 2C_k^2 \int_0^1 (1-u)^k (1-ru)^k du, \quad C_k = \frac{k+1}{2}.$$

3 Polynomial and hypergeometric forms

$$H_k(r) = 2C_k^2 \sum_{j=0}^k (-1)^j \binom{k}{j} B(j+1, k+1) r^j,$$

and

$$H_k(r) = \frac{(k+1)^2}{2} {}_2F_1(-k, 1; k+2; r).$$

4 Derivative identity

$$H_k^{(m)}(r) = (-1)^m 2C_k^2 \frac{k!}{(k-m)!} \int_0^1 u^m (1-u)^k (1-ru)^{k-m} du.$$

Hence

$$(-1)^m H_k^{(m)}(r) \geq 0.$$

5 Endpoint values

$$H_k(1) = \frac{(k+1)^2}{2(2k+1)}, \quad H'_k(1) = -\frac{(k+1)^2}{4(2k+1)}.$$

6 Remarks

This draft intentionally contains only results that were discussed and checked. Higher-dimensional extensions are left for future work.